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CSC 450

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CSC 450 - Assignment #1

1.1 - Suppose the hypothetical processor of Figure 1.3 also has two I/O instructions:

0011 = Load AC from I/O

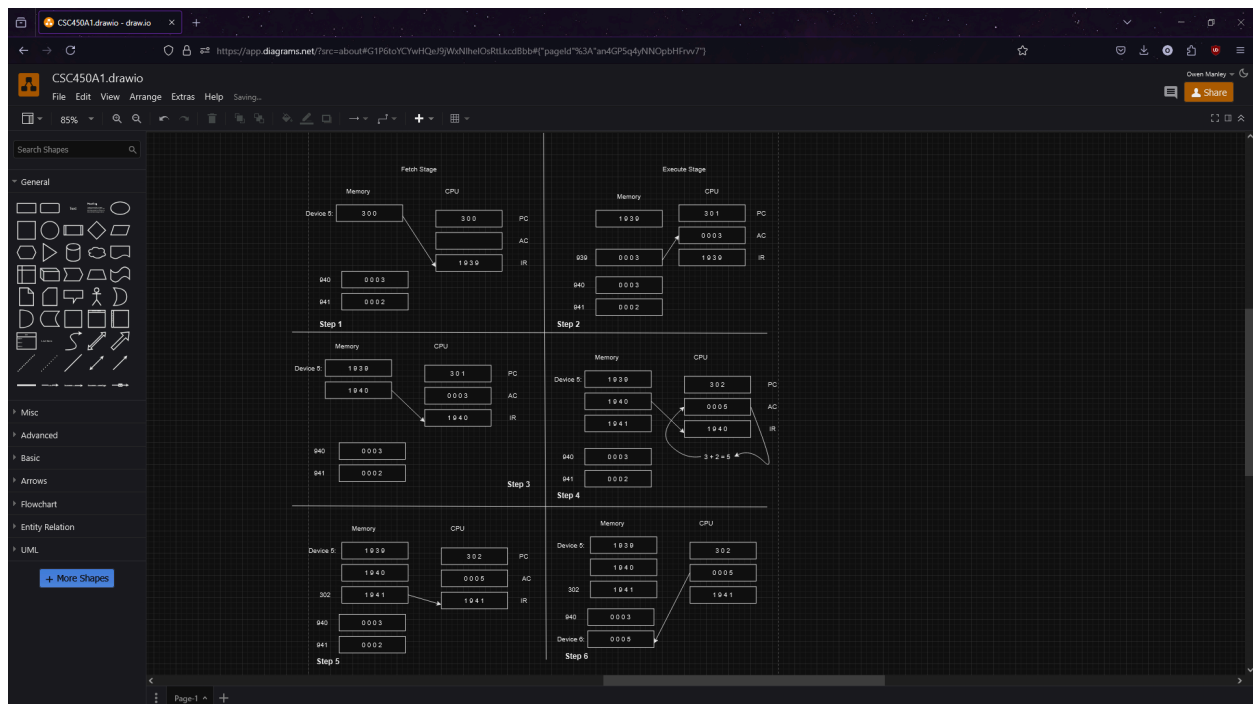
0111 = Store AC from I/O

In these cases, the 12-bit address identifies a particular external device. Show the program execution (using the format of Figure 1.4) for the following program:

Load AC from device 5.

Add contents of memory location 940.

Store AC to device 6.



1.2 - The program execution of Figure 1.4 is described in the text using six steps. Expand this description to show the use of the MAR and MBR.

1: The PC contains 300, the address of the first instruction. This instruction (the value 1940 in hexadecimal) is loaded into the IR and the PC is incremented. MAR is used here to hold the address of the first instruction in temporary memory and is used to hold the next instruction in memory. MBR is then used to transfer the value of the address into the PC and IR.

2: The first 4 bits (first hexadecimal digit) in the IR indicate that the AC is to be loaded from memory. MAR is used here to access the address of the AC, MBR is used to transfer the data into the memory. The remaining 12 bits (three hexadecimal digits) specify the address, which is 940.

3: The next instruction (5941) is fetched from location 301 and the PC is incremented. MAR is used to find location 301, and MBR is used to transfer the address into memory.

4: The old contents of the AC and the contents of location 941 are added, and the result is stored in the AC. MBR makes the calculation and transfers it through MAR which hold it in a temporary address. Then, it is stored in the AC of 941.

5: The next instruction (2941) is fetched from location 302, and the PC is incremented. MAR finds the address and MBR transfers the data from the address to actual memory.

6: The contents of the AC are stored in location 941. MBR transfers the newly calculated data in temporary storage to 941.

1.3 - Consider a hypothetical 32-bit microprocessor having 32-bit instructions composed of two fields. The first byte contains the opcode, and the remainder an immediate operand or an operand address.

a: What is the maximum directly addressable memory capacity (in bytes)? Since the first byte is left to opcode, that means that there is 3 bytes left for operand or operand addresses. With this in mind, the maximum would come out to be 2^{24} memory locations. Which comes out to 16MB.

b: Discuss the impact on the system speed if the microprocessor bus has:

1: 32-bit local address bus and a 16-bit local data bus.

A 32 bit local address bus would be able to hold a much larger memory space at the cost of a reduced execution speed since the local data bus is only 16-bit. Which, reduces the speed to 2 bytes per execution cycle.

2: 16-bit local address bus and a 16-bit local data bus. A 16-bit local address bus would significantly effect memory capacity and wouldn't allow the system to fully utilize its instruction set. It also has the same impact as case 1 with a 16-bit local data bus only allowing for 2 executions per cycle. Overall, a microprocessor with this would be a significant disadvantage.

1.4 - Consider a hypothetical microprocessor generating a 16-bit address (e.g., assume the program counter and the address registers are 16 bits wide) and having a 16-bit data bus.

a: What is the maximum memory address space that the processor can access directly if it is connected to a "16-bit memory"? Since the bus is 16-bit, it can access up to 2^{16} unique addresses. So, $2^{16} * 2$ bytes per cycle = 128KB of memory can be accessed.

b: What is the maximum memory address space that the processor can access directly if it is connected to an "8-bit memory"? Compared to the 16-bit memory, this memory will hold only 1 byte per cycle. So, $2^{16} * 1$ byte per cycle = 64KB of memory can be accessed.

1.10: Consider the following code:

```
for (i = 0; i < 20; i++)  
    for (j = 0; j < 10; j++)  
        a[i] = a[i] * j
```

a: Give one example of the spatial locality in the code.

In this example, $a[i]$ specifies spatial locality because the loop has to iterate through each index. Since it is the outer loop, they are physically the closest to each other in the cache.

b: Give one example of the temporal locality in the code.

In this example, $a[0]$ - $a[9]$ are likely accessed multiple times due to the calculation involving the inner loop. Because of the calculation, each index is accessed multiple times again to perform the calculation.

1.12 - Consider a memory system with the following parameters:

$T_c = 100 \text{ ns}$ $C_c = 0.01 \text{ cents/bit}$

$T_m = 1,200 \text{ ns}$ $C_m = 0.001 \text{ cents/bit}$

a: What is the cost of 1 MByte of main memory?

$1\text{MB} = 2^{20} * 8 = 8,388,608 \text{ bits}$

$\text{Total cost} = 8,388,608 * 0.001 = 8,388.608 = \83.89

b: What is the cost of 1 MByte of main memory using cache memory technology?

$\text{Cost of 1MByte of main memory using cache technology} = 83,886,08 * 0.01 = \838.86

c: If the effective access time is 10% greater than the cache access time, what is the hit ratio

H? Since T_e is greater than T_c , $T_e = 1.1 = 1.1 * 100 = 110 \text{ ns}$.

$$T_e = H * T_c + (1 - H) * T_m$$

$$110 = 100H + 1200 - 1200H$$

$$110 = 1200 - 1100H$$

$$1100H = 1200 - 110$$

$$1100H = 1090$$

$$H = 1090/1000 = 0.99\%$$